# EART 27201 Week 6

Carbonate rocks
(but mostly about limestones)

# Kinds of Sedimentary Rocks

## Clastic (detrital): Physical Processes

allochthonous

Terrigenous: siliciclastic rocks (consist of fragments of pre-existing rocks, transported and deposited by physical processes e.g. cg, ss, ms)

Penecontemporaneous volcanic activity: volcaniclastics (consist of lava and rock fragments e.g. tuffs, ignimbrites etc.)

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## Non-clastic: Chemical and Biological Processes

autochthonous

Biogenic, biochemical and organic processes: e.g limestone, coal, oil shale and chert (oozes)

Chemical processes: evaporites, flints and ironstones

# Limestones – the most important rocks…?

- Contain at least 50% of world's hydrocarbon reserves
- Host major metallic ore deposits
- Aquifers
- Raw material for chemical industry
- Fertiliser
- Aggregate
- Cement, concrete
- Storehouses of $CO_2$ (but only temporary- why?….)



A limestone overlying a succession of poorly consolidated mudstones

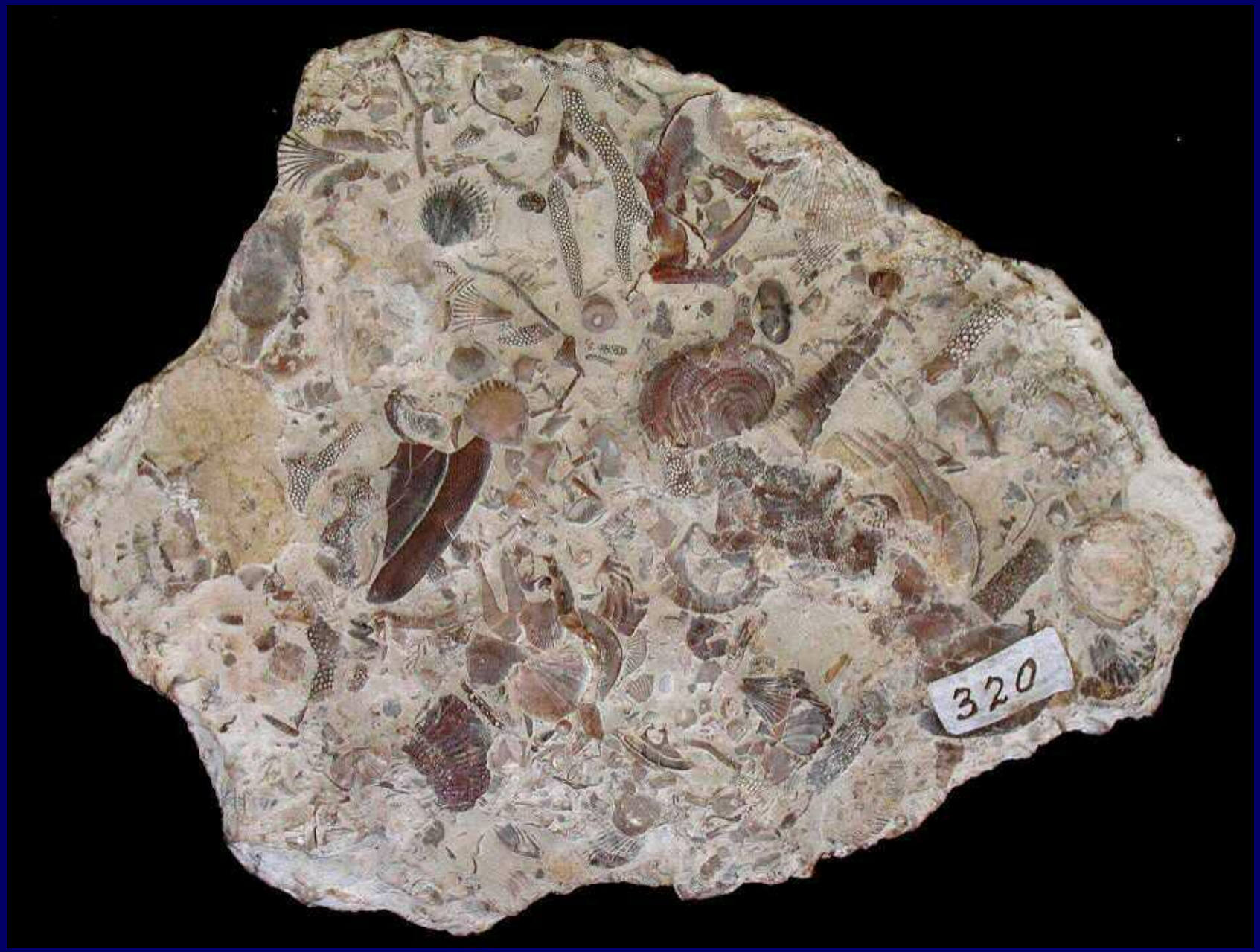


Shallowing-upwards successions

If carbonate production and accumulation may exceed the rate of subsidence or sea-level rise, sediments can build up to sea-level
Results in  a vertical succession of rock types in which each unit is deposited in successively shallow water



The location of carbonates today has little to do with their location at the time of formation




The marine origin of limestones is indicated by its constituent parts

Many limestones are made up of whole and fragmented fossils, in this case mostly crinoids, brachiopods and bryozoans

# Contrasting carbonates with terrigenous clastics

- Carbonates

- Mostly occur in shallow tropical marine environments, because both chemical and biological production more rapid in warm water

- Terrigenous clastics

- Marine, non-marine, climate no constraint, just need sediment supply and hole to put it in

# Contrasting carbonates with terrigenous clastics

- Carbonates
- Grain size may reflect size of calcified organism hardparts

- Terrigenous clastics
- Grain size reflects wave and current energy

# Contrasting carbonates with terrigenous clastics

- Carbonates

- Carbonate mud produced in situ. Presence may indicate prolific growth of organisms whose calcified parts are mud-sized crystals e.g. calcareous algae

- Terrigenous clastics

- Terrigenous mud produced by chemical weathering. Deposition may involve settling out of suspension

# Contrasting carbonates with terrigenous clastics

- Carbonates

- Organic structures such as reefs can buildup and modify hydraulic conditions in surrounding environments over long timescales

- Terrigenous clastics

- Bed configurations and bedforms created by wave and current and wind energy do affect flow conditions but these are mobile (relatively short-lived effect)

# Contrasting carbonates with terrigenous clastics

- Carbonates

- Sediments may be cemented on the sea floor producing lithified surfaces (hardgrounds) or make rigid cemented structures like reefs

- Terrigenous clastics

- Sediments normally remain uncemented in environment of deposition

# Contrasting carbonates with terrigenous clastics

## Carbonates

- Sealevel fall and subaerial exposure lead to intensive diagenesis and strong modification of depositional texture

## Terrigenous clastics

- Sealevel fall and subaerial exposure leaves sediments more or less unaffected



An exhumed palaeokarstic (potholed) surface exposed on the foreshore by marine erosion.

# Contrasting carbonates with terrigenous clastics

- Carbonates
- Characteristic features often obliterated by low grade metamorphism

- Terrigenous clastics
- Characteristic features e.g. grain size and sedimentary structures mostly survive low grade metamorphism e.g. Dalradian

# Carbonate Rocks

The carbonate rocks make up 10 to 20% of sedimentary rocks.

Mostly marine in origin (some lacustrine carbonates)

They largely consist of two types of rocks:

- Dolostones which are composed mostly of dolomite [$CaMg(CO_3)_2$]

- Limestones which are composed mostly of calcite ($CaCO_3$) or high Mg calcite [$(Ca,Mg)CO_3$]

# Carbonate Precipitation

Modern sediments are made of two minerals, both $CaCO_3$: **Calcite** and **Aragonite**

Limestones often described as “chemically precipitated” but…

Calcium carbonate removed from seawater in *two* ways:

- Chemically (inorganically)
- Biochemically

Chemically (inorganically), by direct precipitation from supersaturated water

- fine crystals in the water column, which then settle to the seafloor, or as successive spherical shells deposited around a nucleus particle
- although calcite more stable in pure water, observed arganonite is precipitated from sea water

# Carbonate Precipitation

Most however,

Biochemically, by organisms to construct skeletons

- calcareous algae: *Halimeda* and other green algae, aragonite; *Lithothamnium* and other reef-making red algae, calcite
- modern corals (hexacorals): aragonite
- Paleozoic corals (rugose corals, tabulate corals): probably calcite, by good preservation of structures
- brachiopods, bryozoans, foraminifera: calcite then and now
- echinoderms: calcite then and now; large single-crystal skeletal components, very durable
- trilobites: probably calcite, by good preservation of structures
- mollusks: mostly aragonite, sometimes partly or entirely calcite

Aragonite is metastable and is not preserved in lithified limestones

*Originally* aragonite grains preserved as moulds or as calcite casts

*Originally* calcite grains preserved with original structures intact

# Controls on Carbonate Precipitation

Reaction for precipitation of $CaCO_3$:

$$CaCO_3\ (solid) + H_2O + CO_2 \Leftrightarrow Ca^{2+} + 2CO_3^{2-} + 2H^+$$

So
*anything that **increases** the concentration of dissolved $CO^2$ tends to cause **dissolution** of calcium carbonate,*

and
*anything that **decreases** the concentration of dissolved $CO^2$ tends to cause **precipitation** of calcium carbonate.*

Important effects
temperature*:* as the water temperature increases, the equilibrium solubility of $CO^2$ decreases, so as sea water is warmed there is a tendency for $CO^2$ to be released back into the atmosphere and for $CaCO^3$ to be precipitated.

photosynthesis*:* in photosynthesis, plants take up $CO^2$ from the environment and fix it in organic compounds in their tissues, thereby releasing oxygen.

where organisms live: carbonate-secreting marine invertebrates live in greatest numbers in the warm shallow parts of the oceans. Can secrete $CaCO_3$ even from water that is not saturated in $CaCO_3$, but they do it best and most abundantly where the water is saturated

# Limestones

As with terrigenous clastic sediments, limestones can be considered as comprising grains, matrix and cement

Grains = allochemical particles or allochems

Four common types:

Bioclasts (plants and animals)

Ooids
Peloids
Intraclasts

} Non-biogenic constituents

See practical handout



Many limestones are composed of bioclasts - whole or fragmented shells and other organism hard parts

# Bioclasts: Skeletal fragments

Whole or broken skeletons of organisms (fossils) e.g., gastropods, crinoids, bivalves, corals and plants (algae)

Precipitated by organisms that formed elsewhere and became included in the carbonate sediment

Range in size from gravel to fine sand

Relatively short distance of transport (not resistant)

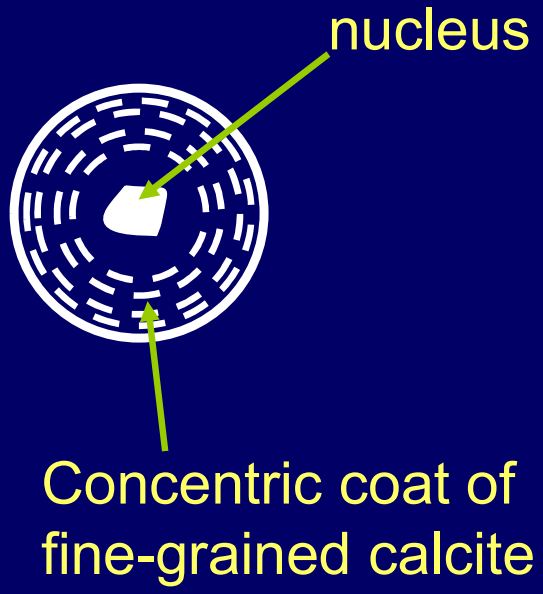
Degree of rounding and sorting of the grains may not be a reflection of the energy of the transporting medium, but may be biologically determined

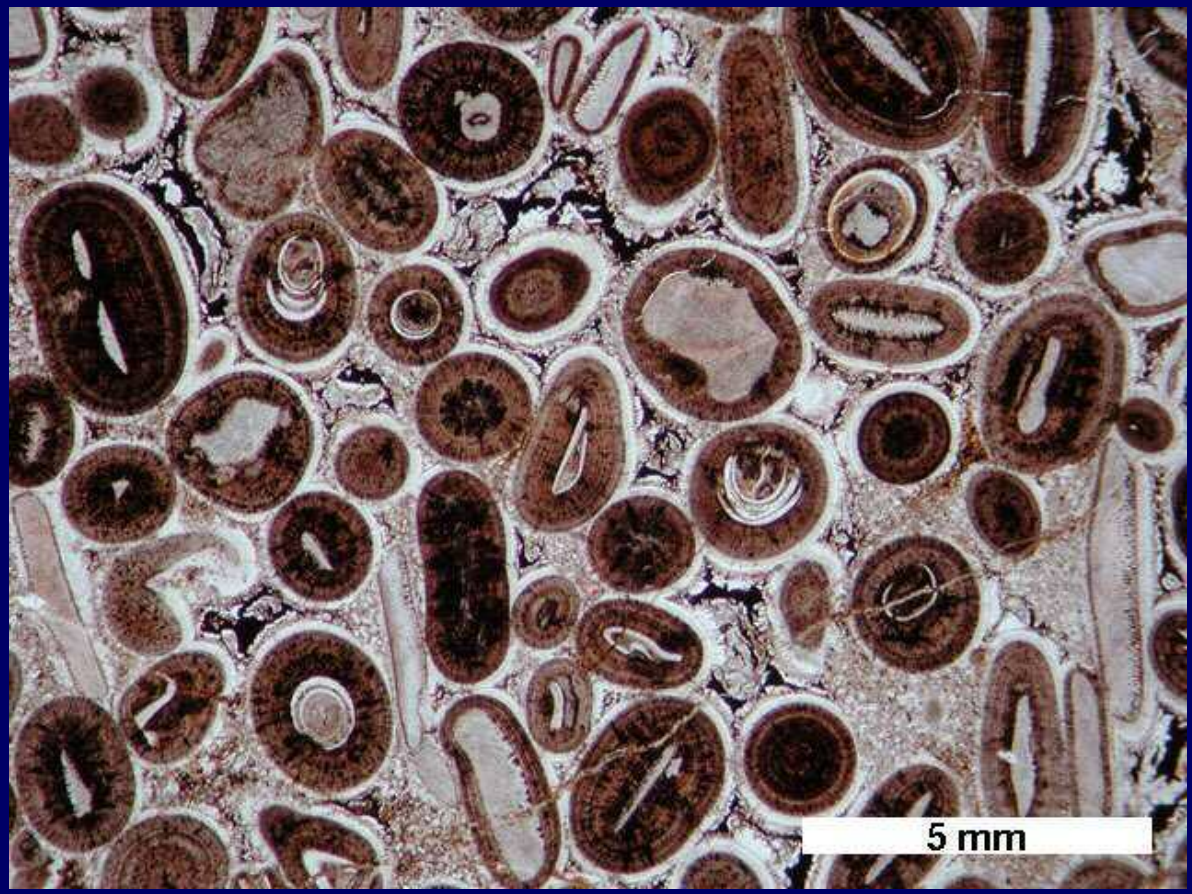
- inherited rounded shape
- well sorted if many of the same size organisms die at the same place

# Ooids

Spherical or ellipsoidal bodies of $CaCo_3$

- < 2mm in diameter (>2mm in diameter 'Pisoids')
- internal structure of concentric layers
- central nucleus (fragment of carbonate material or clastic sand grain)



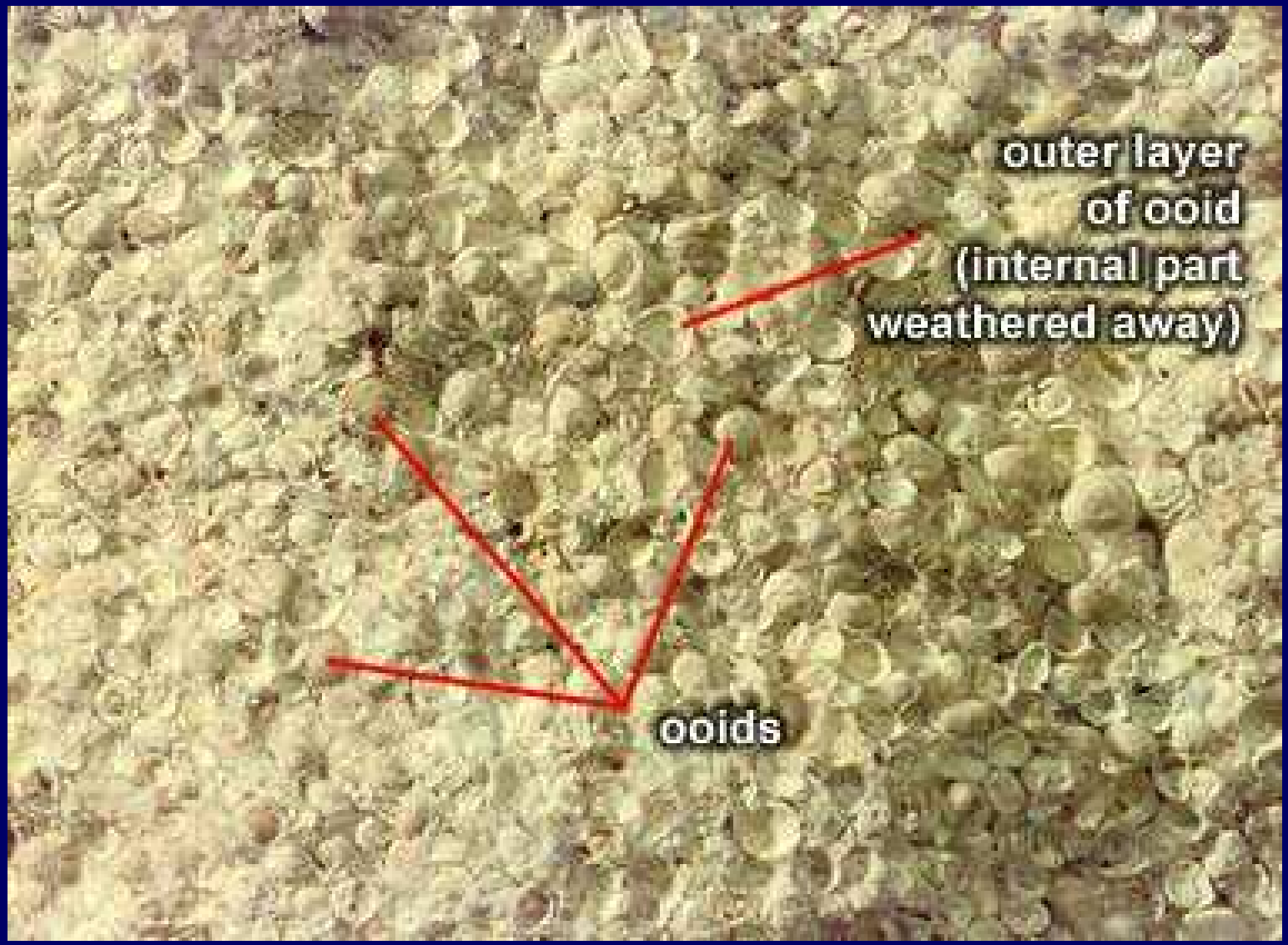


Modern ooid sand from the Bahamas.  Most grains 0.25-0.5mm.  Form shoals in shallow marine environments today.



Oolitic limestone.  Ooids average 0.5mm in diameter.

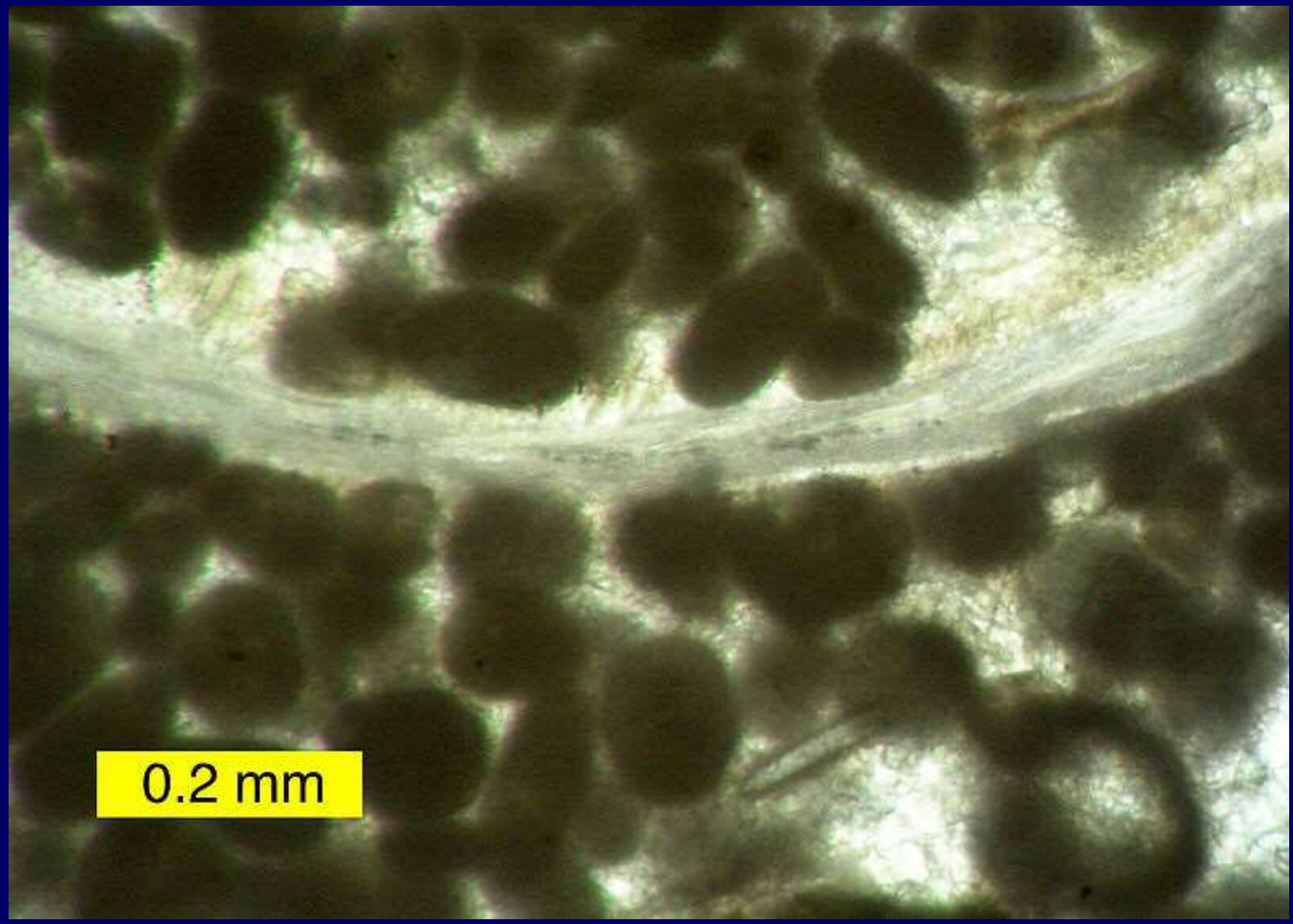
Component of limestones throughout the Phanerozoic




Ooids in close up. The concentric structure of the oolitic coating is revealed in the way the grains break. Ooids 0.5-1.0mm in diameter. Origin has been subject of debate. Form by chemical precipitation out of agitated water saturated in CaCo3 in warm waters.

# Peloids

No concentric structure
Rounded, structureless, fine-grained carbonate
Mostly <1 mm diameter
Faecal pellets of marine organisms e.g., gastropods



# Matrix and cement in limestones

- Matrix is carbonate mud (carbonate finer than 0.063mm). No one source?
- Very finest material may be a chemical precipitate (micrite = clay-sized carbonate)
- Coarser material can be from abrasion of larger grains
- Much carbonate mud formed from disintegration of encrustations around calcareous algae. These plants photosynthesize and calcium carbonate is precipitated:

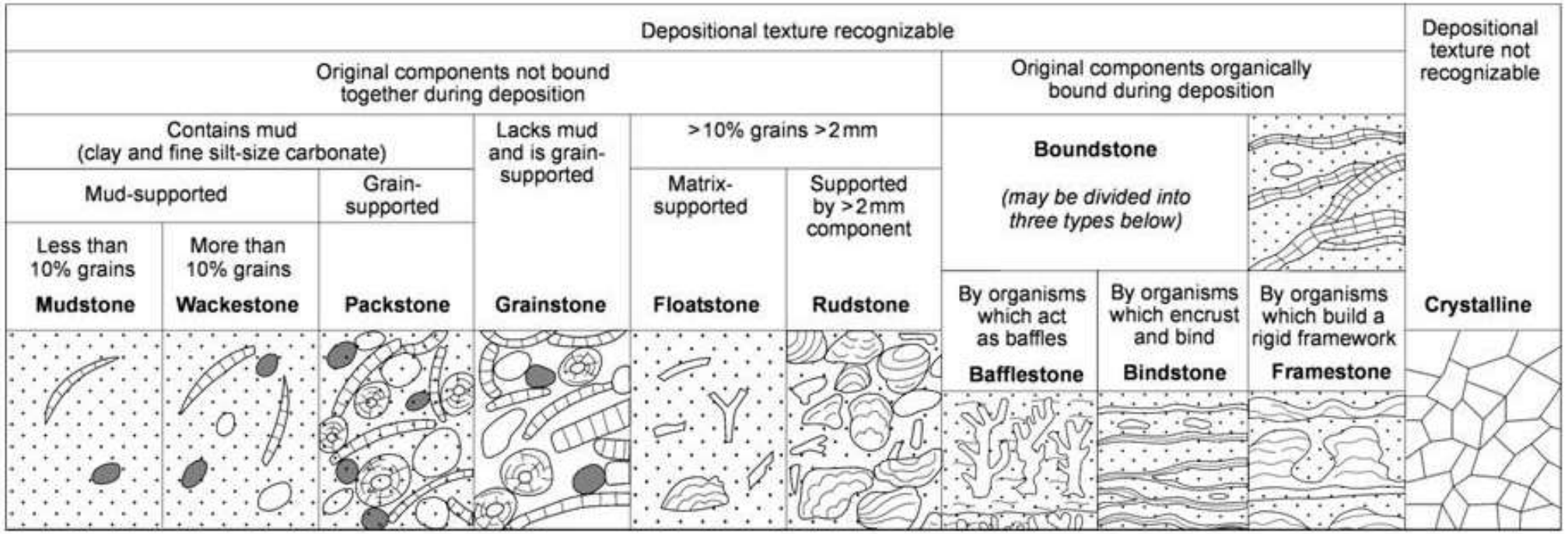
$$Ca^{2+} + 2HCO_3^- = CaCO_3 + H_20 + CO_2$$

- Cement is calcite precipitated into space after deposition, producing crystalline calcite known as sparite

# Classification of Limestones

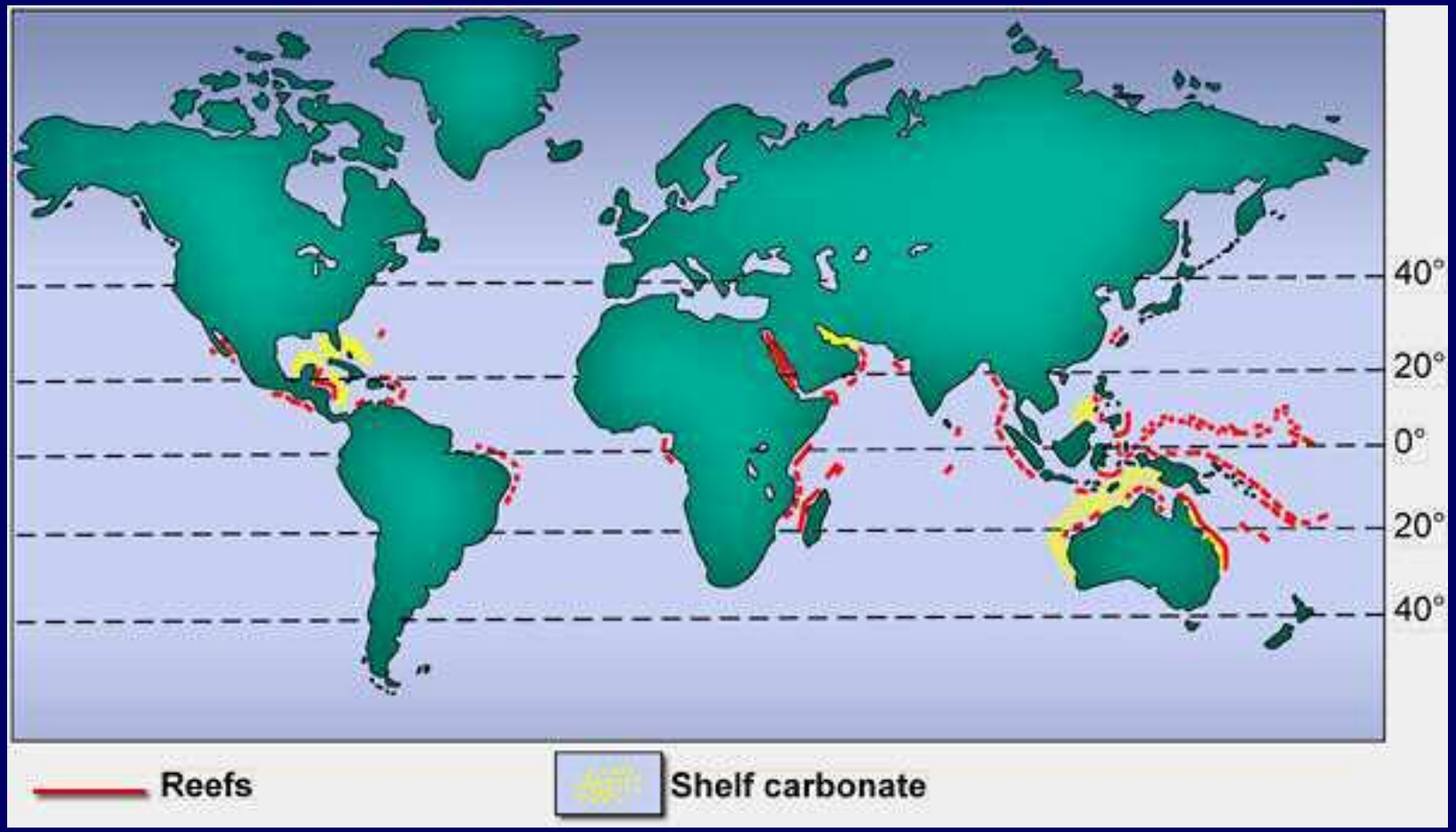
- A grain size classification can be useful in the field
  - Average size >2mm = Calcirudite
  - Average size 0.063 – 2mm = Calcarenite
  - Average size <0.063mm = Carbonate Mudstone
- Most popular classification is Dunham's classification according to depositional texture. Often difficult to apply without microscopic study.

# Dunham's Classification Scheme



# Deposition of Carbonate Sediments

In modern environments, carbonates are deposited in both shallow and deep waters

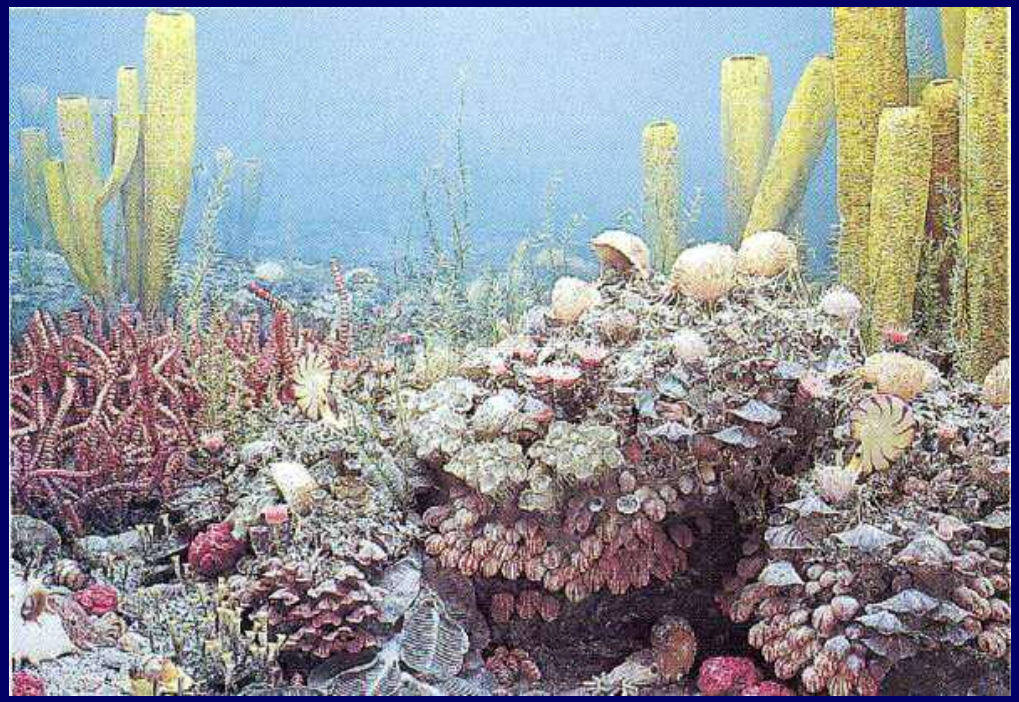



Shallow water areas are relatively small e.g. parts of Caribbean, Arabian Gulf and reefs of Indian and Pacific Oceans

# Deposition of shallow water carbonate sediments

These are areas free from terrigenous clastic input

Clear waters needed for growth of larger organisms such as corals



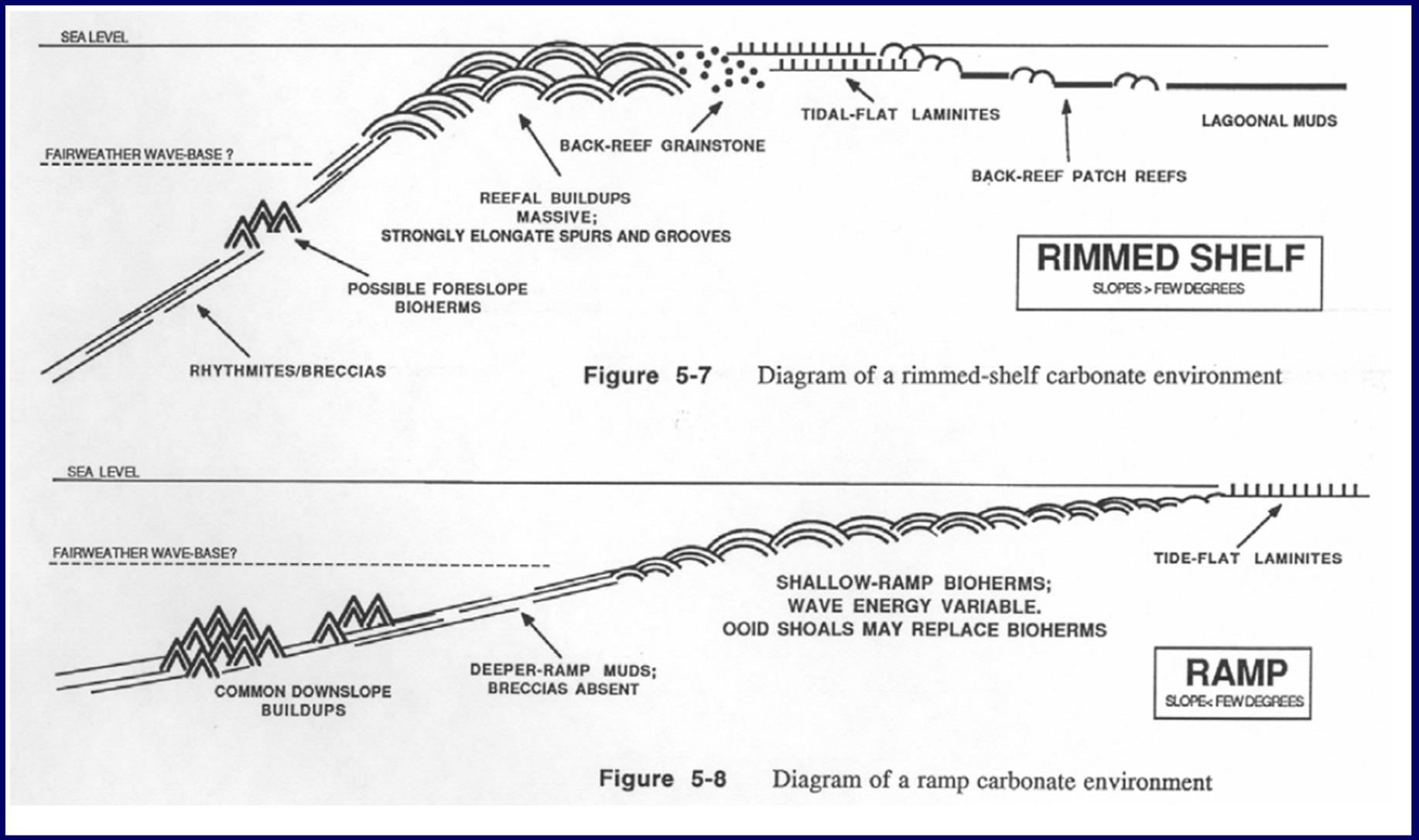
A modern reef comprised of larger carbonate-secreting organisms such as corals



The Bahama Banks

Warm waters promote both biological production and chemical precipitation of carbonate

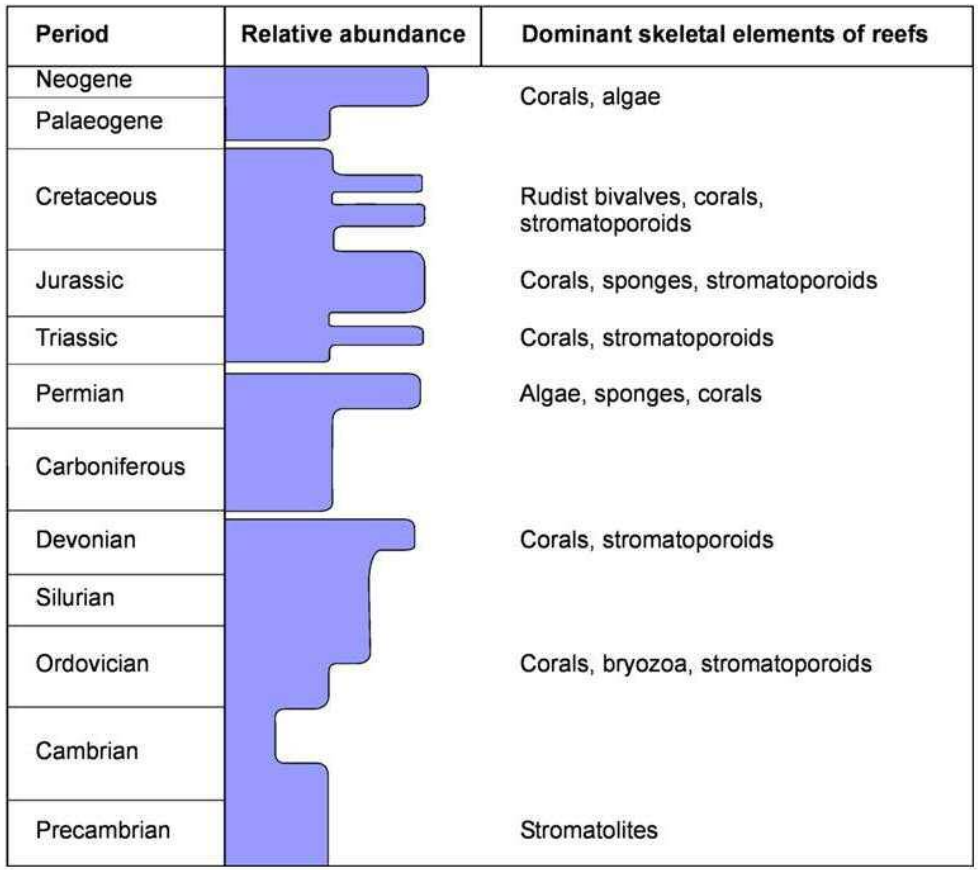
# Deposition of shallow water carbonate sediments



# Changes in main reef builders over geologic time

In geological record, times when extensive areas of shallow water carbonate deposition much more common than today

**Type and abundance of carbonate reefs through time**



| Period | Relative abundance | Dominant skeletal elements of reefs |
|---|---|---|
| Neogene | | Corals, algae |
| Palaeogene | | |
| Cretaceous | | Rudist bivalves, corals, stromatoporoids |
| Jurassic | | Corals, sponges, stromatoporoids |
| Triassic | | Corals, stromatoporoids |
| Permian | | Algae, sponges, corals |
| Carboniferous | | |
| Devonian | | Corals, stromatoporoids |
| Silurian | | |
| Ordovician | | Corals, bryozoa, stromatoporoids |
| Cambrian | | |
| Precambrian | | Stromatolites |

(Nicholls, 2009)

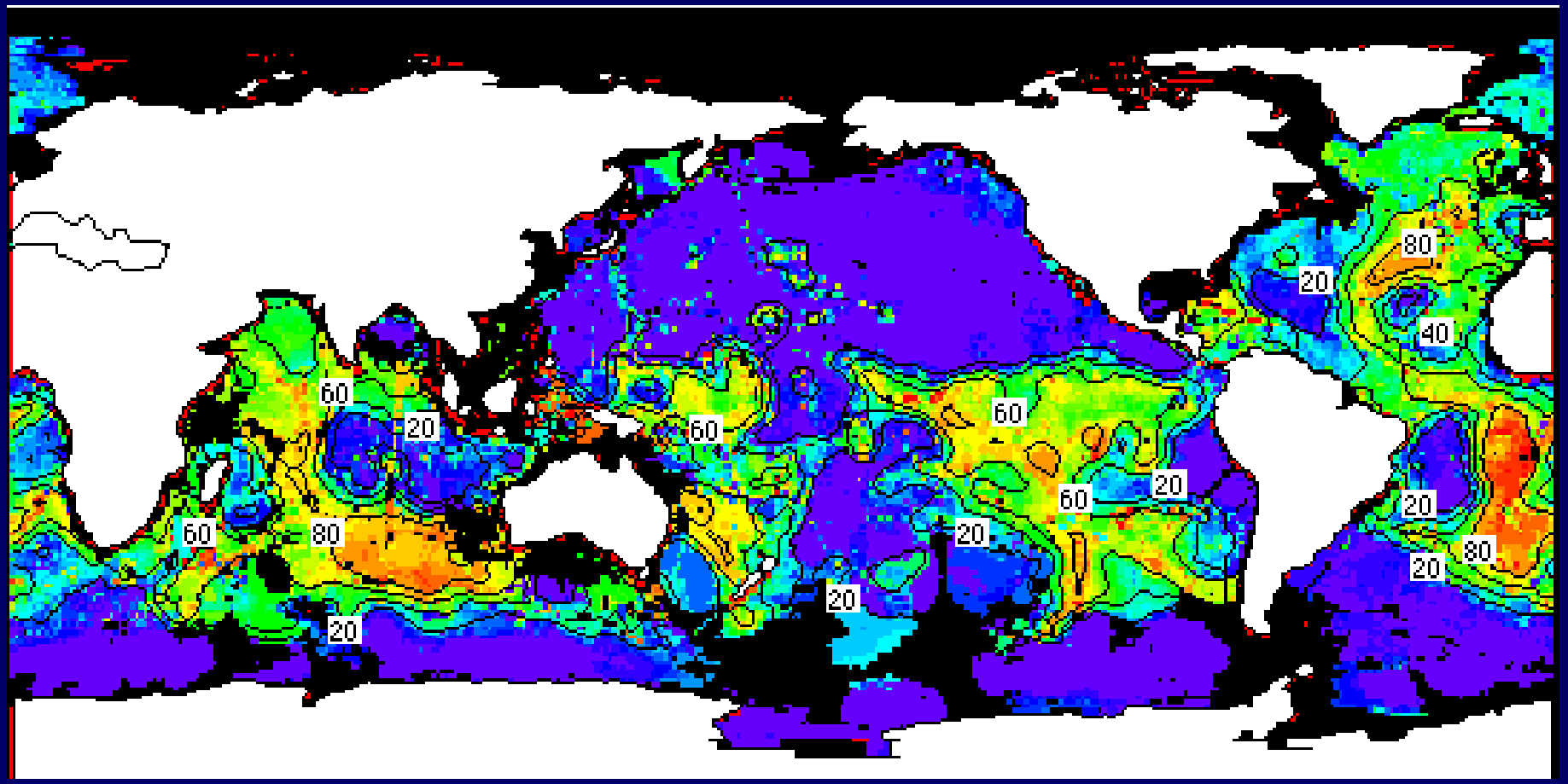
Today global sea-level is low and continents stand high

- reduced areas of shallow water
- increased terrigenous input

# Deposition of deep water carbonate sediments

Carbonate sediments (pelagic carbonates) also found over large areas of deep ocean floor (depths >50-100 m)

Primarily in tropical regions, between 30°N and 30°S




%$CaCO_3$ in surface sediments (Archer, 2000)

# Modern Pelagic Sediments

Deposition dominated by settling out of suspension = Pelagic Sediments

Because of depth skeletal remains of sea-floor dwellers not a significant component
Biogenic portions of these sediments are plankton

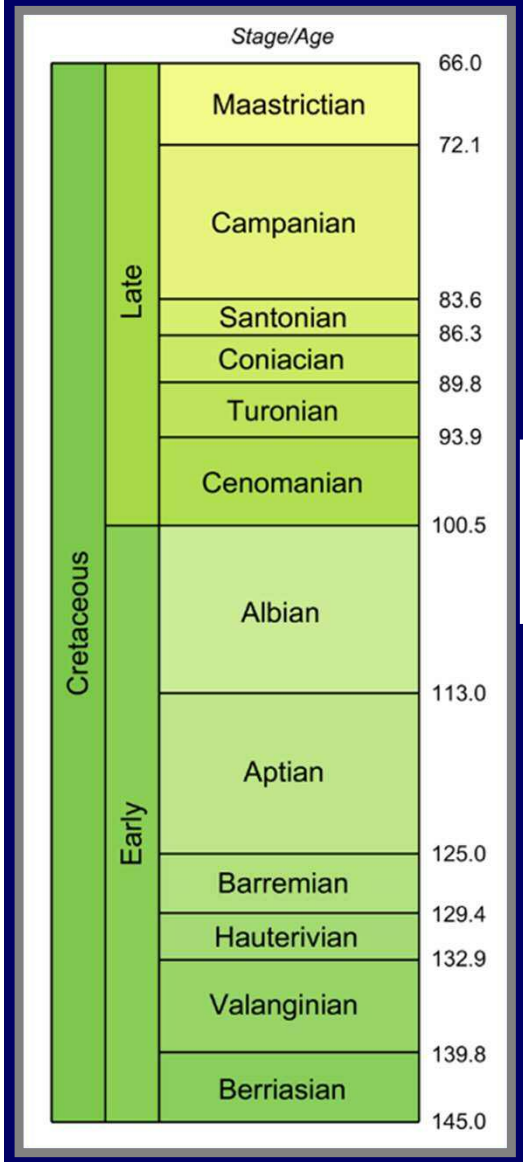
Includes both plants and animals

- Plants with calcite skeletons - Coccolithophorids
- Plants with silica skeletons e.g. diatoms
- Animals with calcite skeletons – planktonic Foraminifera
- Animals with aragonite skeletons – Pteropods (small pelagic gastropods)
- Animals with silica skeletons - Radiolaria

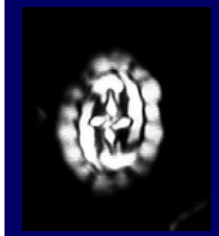


A selection of planktonic foraminiferans, typically less than 0.25mm across (Foraminiferal ooze most abundant ooze today)

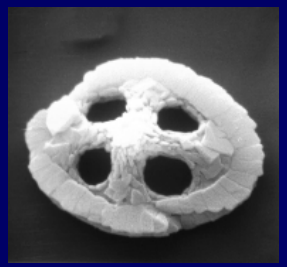
# Deposition of deep water carbonate sediments

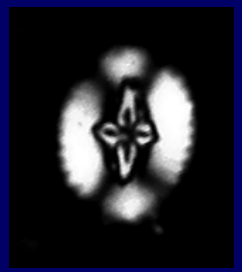


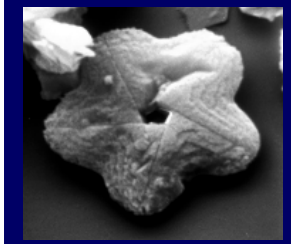

Biostratigraphic range

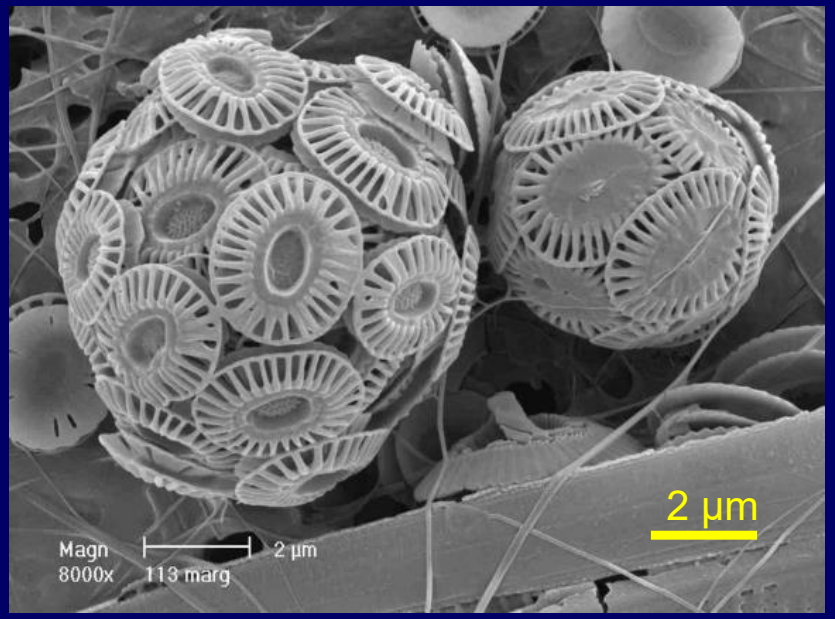













SEM photograph of *Emiliyana huxleyi* (a modern coccolithophore). Note the overlapping plates (coccoliths) each about 0.002 to 0.004mm across

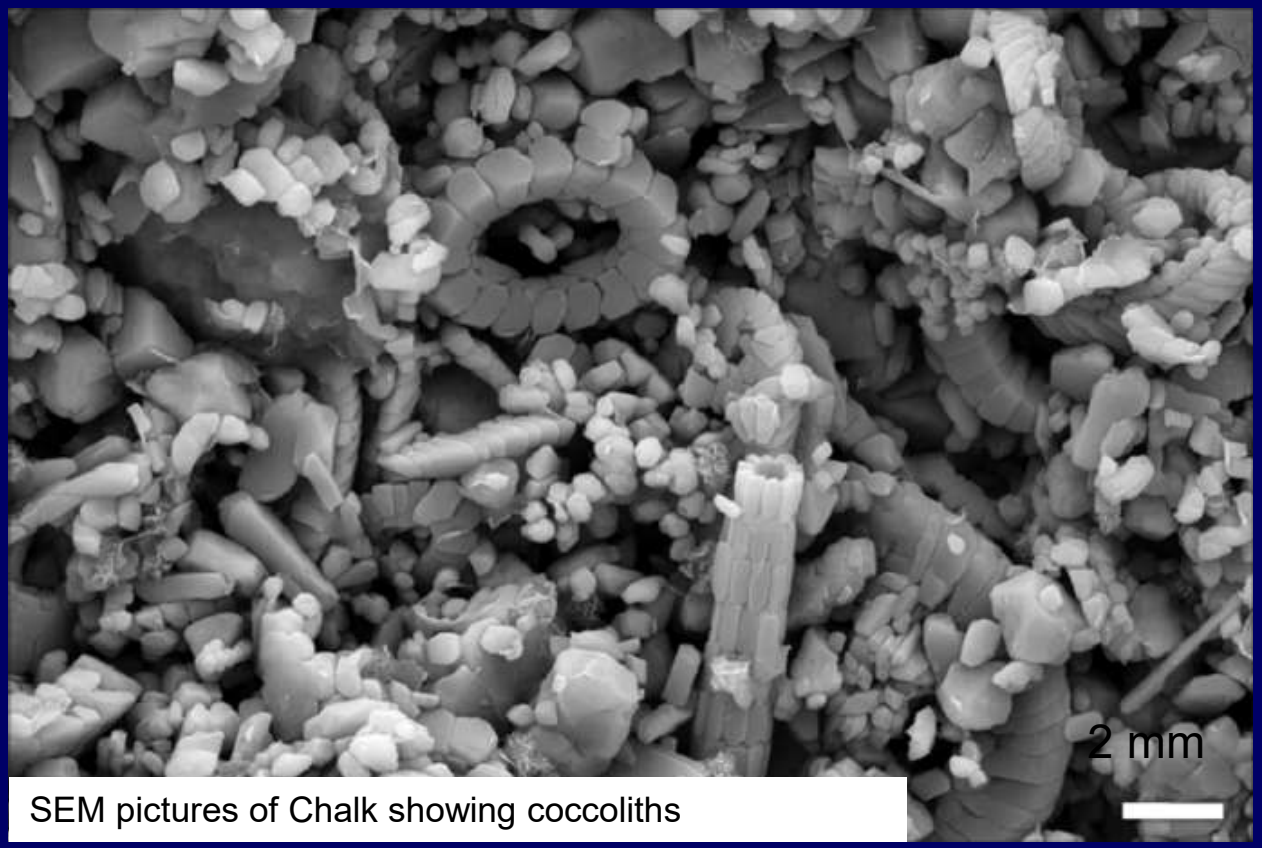
## Coccolithophores

- first appeared after Carnian Pluvial Episode (late Triassic) in epicontinental seas
- consist of an algal cell surrounded by calcite plates called coccoliths

Over half carbonate removed from ocean today is extracted by coccolithophorids

Abundant floating in near surface water of the oceans (planktonic inhabiting the pelagic zone)

# Formation of coccolith ooze and chalk



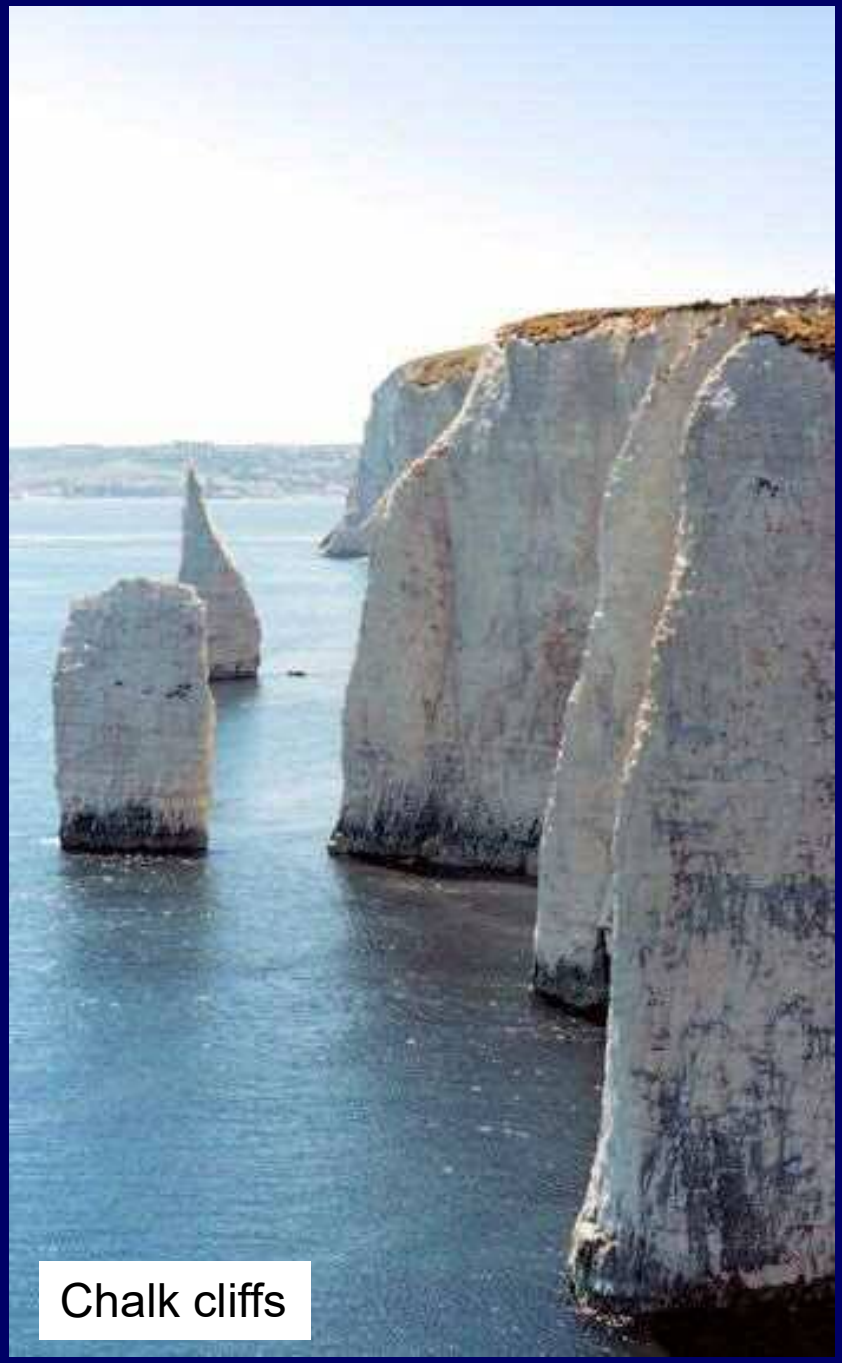

SEM pictures of Chalk showing coccoliths

- When the coccolithophore dies, the sphere breaks apart and coccoliths accumulate on the ocean floor
- Coccolith ooze on the ocean floor lithifies to chalk
- Coccolithophores are a major food source for herbivorous plankton whose faecal pellets are deposited in pelagic sediments
- Influence the global carbon cycle because $CO_2$ is effectively removed from the atmosphere and buried on the sea floor

# Ancient Pelagic Sediments: Cretaceous chalk

Ancient pelagic sediments sometimes accumulated in shallower water than those of today

- Classic case is Upper Cretaceous Chalk, a cemented coccolith ooze with a diverse benthonic fauna including molluscs and echinoids
- Accumulated in water at most a couple of hundred metres deep
- Cretaceous sea-levels very high, reduced terrigenous input, pelagic facies spread into shallower water



Chalk cliffs

# Ancient Pelagic Sediments: Differences

Older pelagic sediments differ from modern sediments because the nature of plankton changes:

- Planktonic forams and coccolithophorids only become important in Cretaceous
- In Triassic and early Jurassic there were some pelagic bivalves and there were larger organisms such as ammonites and belemnites
- In Palaeozoic calcareous plankton almost absent
- Radiolaria have been around throughout Phanerozoic
- Pelagic sediments of Palaeozoic and later times below CCD dominated by Radiolarian Cherts
- Radiolaria made of opaline silica $SiO_2.nH_2O$
- This eventually recrystallizes to quartz siliceous ooze



Diatoms



Radiolaria up to 0.5mm across

# Porosity in Sedimentary Rocks

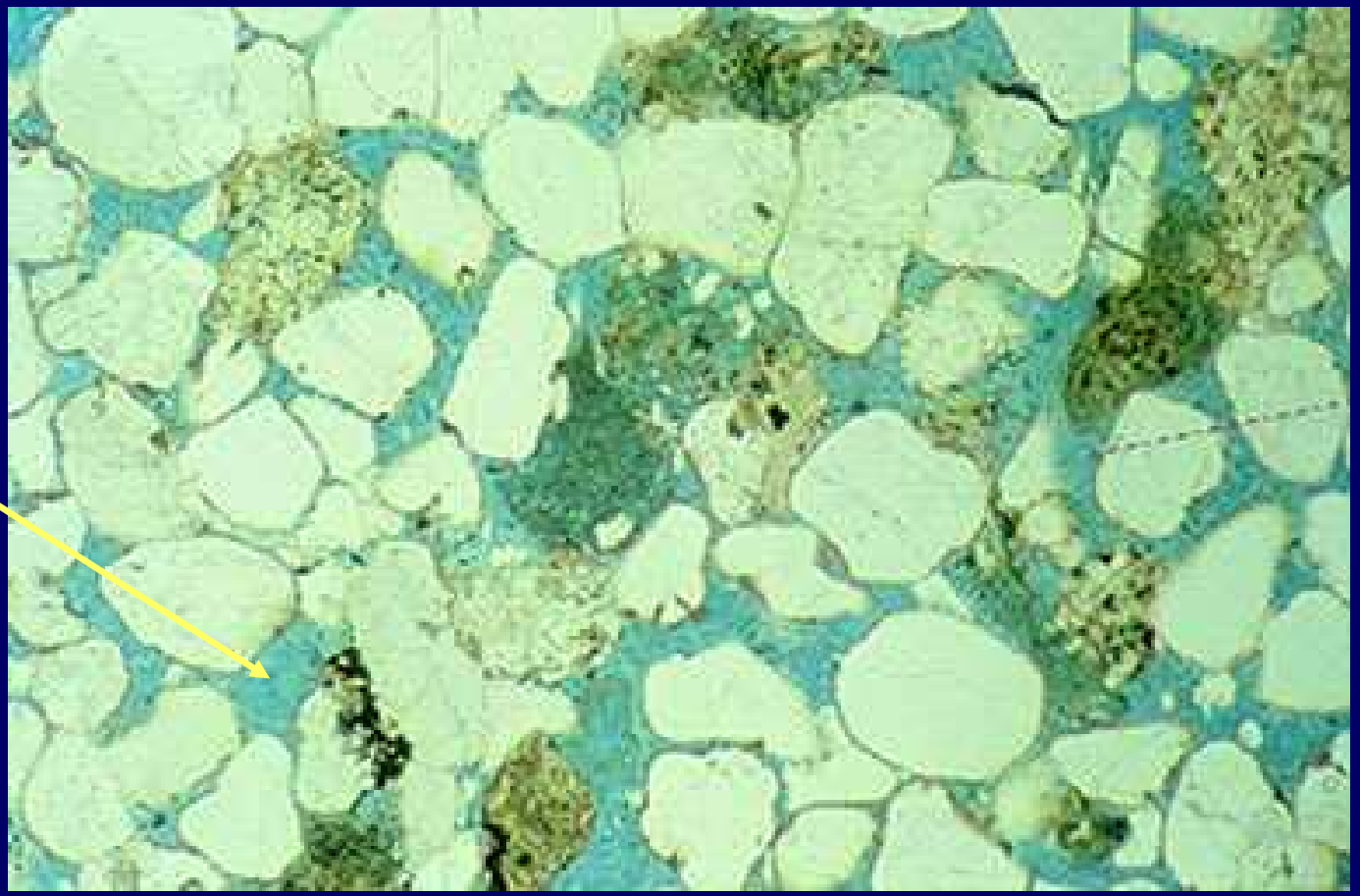
Porosity is the amount of void space, expressed as volume %.

- may be filled with air, water or, occasionally, hydrocarbon
- some porosity is primary – formed at the time of deposition
- other porosity is secondary – it has appeared in the rock as a result of some post-depositional process

# Primary Porosity

Two types:

- intergranular: space between grains
- intragranular: voids within grains e.g., chambered organisms



Pores are filled with blue-dye impregnated resin

Width of field of view approx 3mm

Thin section of a sandstone showing primary intergranular porosity and secondary porosity from solution off some grains (feldspars?)

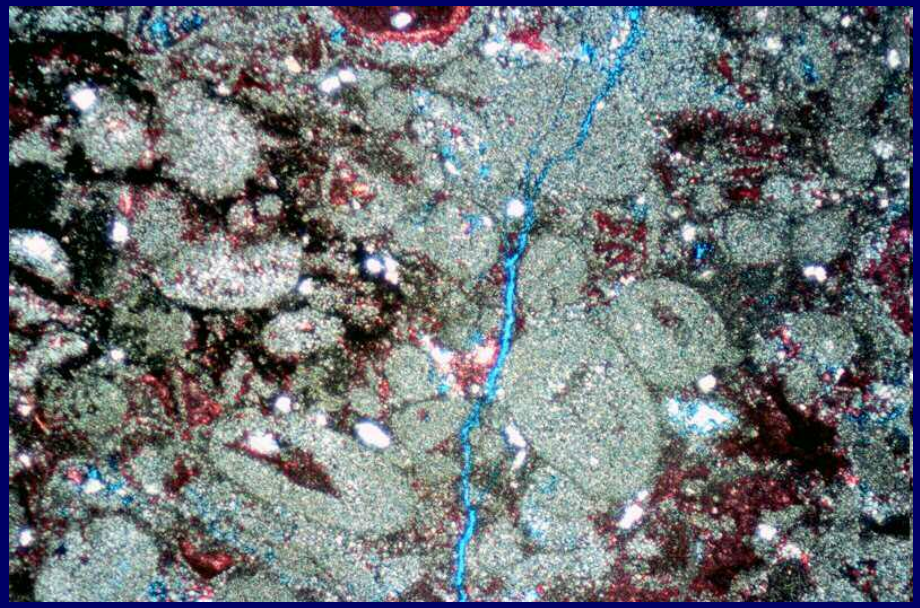
# Secondary Porosity

Secondary porosity develops as a result of:

- dissolution (and dolomitization in limestones)
- tectonic processes e.g. fracturing



Large scale secondary porosity- caverns formed during solution by meteoric water



Fracture porosity (blue filling facture)

# Permeability

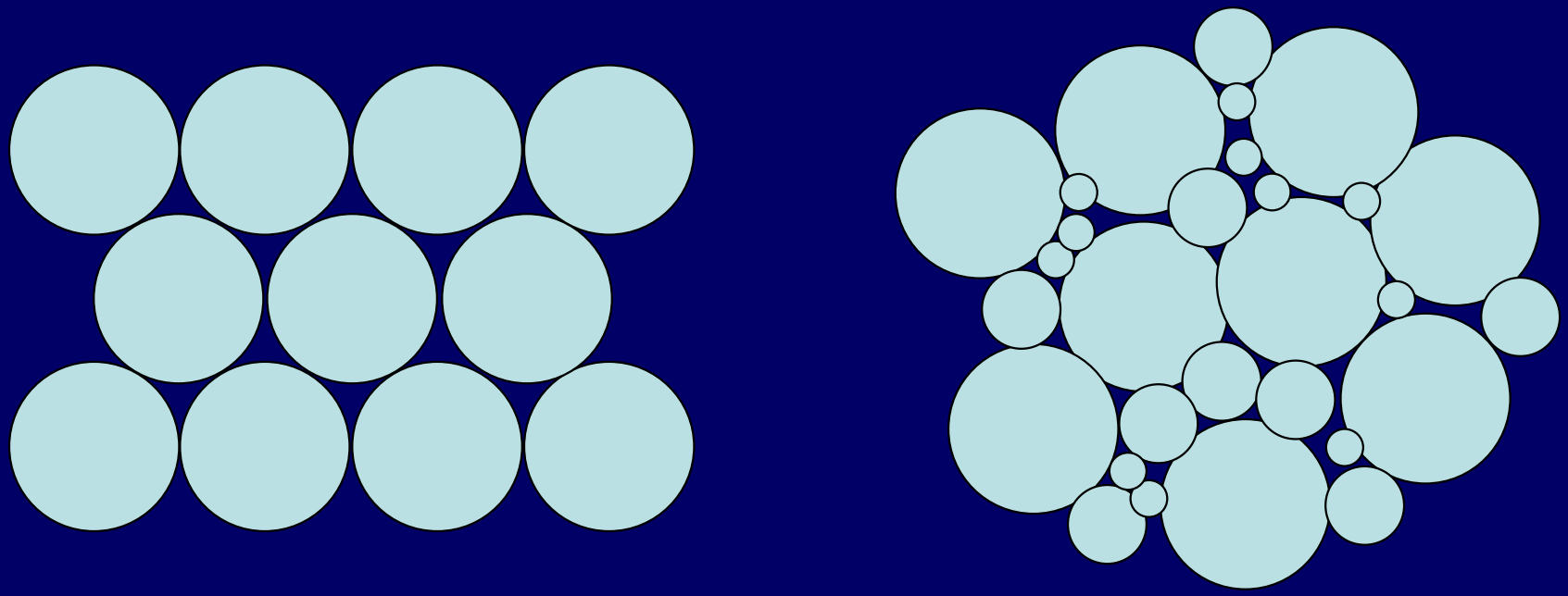
Permeability is the ability of a sediment/rock to allow fluids to pass through it

Depends on:

- the shape and size of pores
- connectivity of pores (effective porosity)
- properties of the fluid

# Relationship between porosity and permeability

(i) Permeable rocks must be porous



Very well sorted

Poorly sorted

(ii) For sandstones, permeability increases as porosity increases (and vice versa)

(iii) Porous rocks need not be permeable!
e.g., wackes vs arenites

# Diagenesis: turning sediment into rocks

Diagenesis: the processes which affect a sediment after deposition

- results in lithification of sediment
- excludes metamorphism (>150-200$^0$C)
- includes burial to depths of several km

Early diagenetic events:
- from sedimentation to shallow burial

Late diagenetic events:
- occur during deep burial and subsequent uplift

Involve both physical and chemical processes

# Diagenesis: turning sediment into rocks

Diagenetic processes may lead to porosity reduction
e.g., cementation, compaction and pressure dissolution

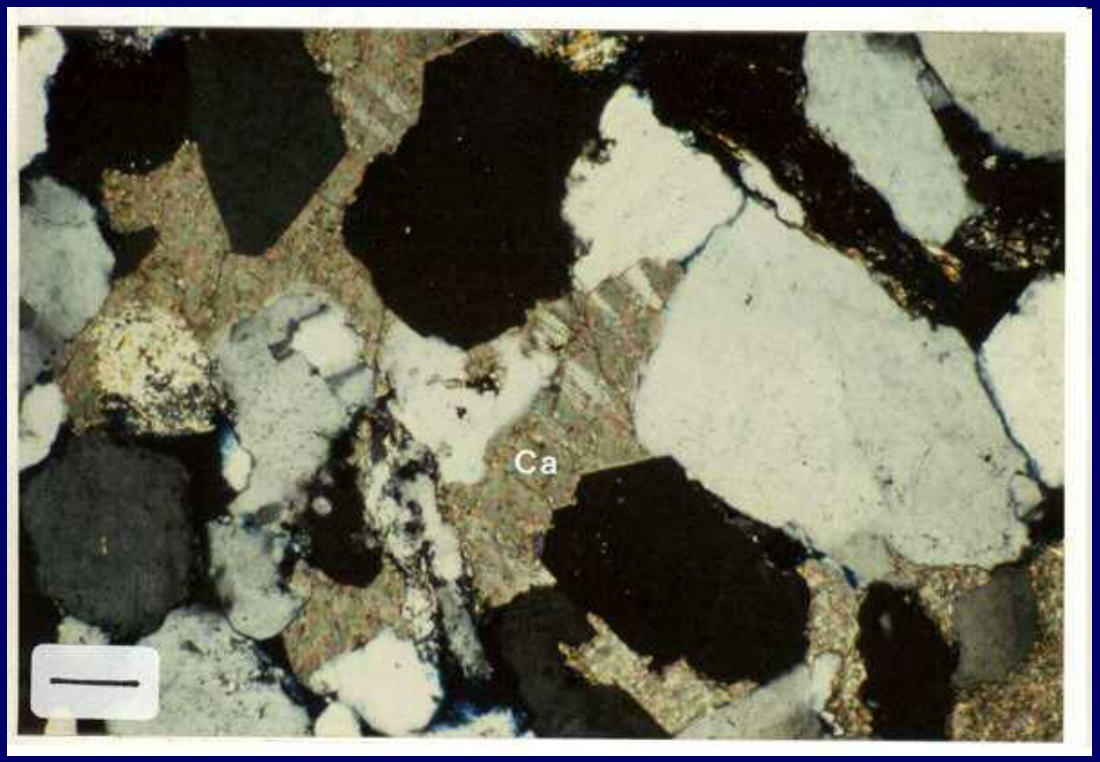
Occasionally processes lead to porosity increase
e.g., dissolution

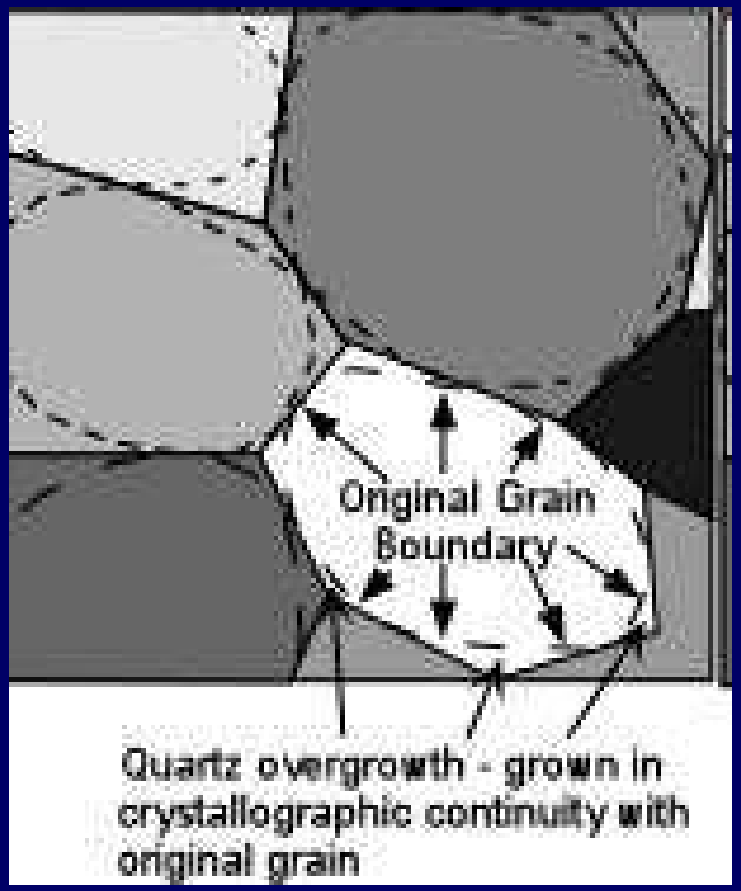
Diagenetic processes can result in the formation of new minerals
e.g., dolomitization in limestones

# Diagenetic processes: Cementation

Cementation is process of precipitation of space filling crystals

- Quartz or calcite in terrigenous clastic rocks
- Calcite in limestones



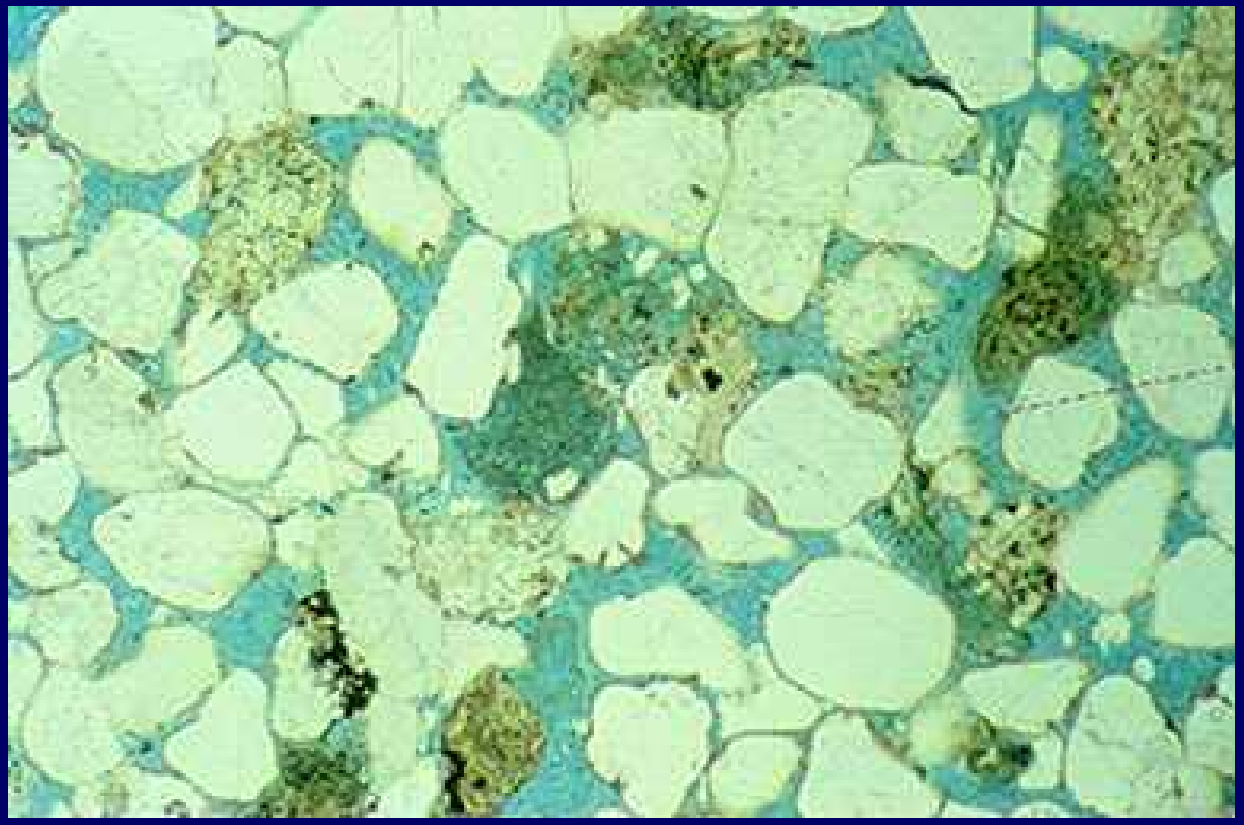



# Diagenetic processes: Compaction

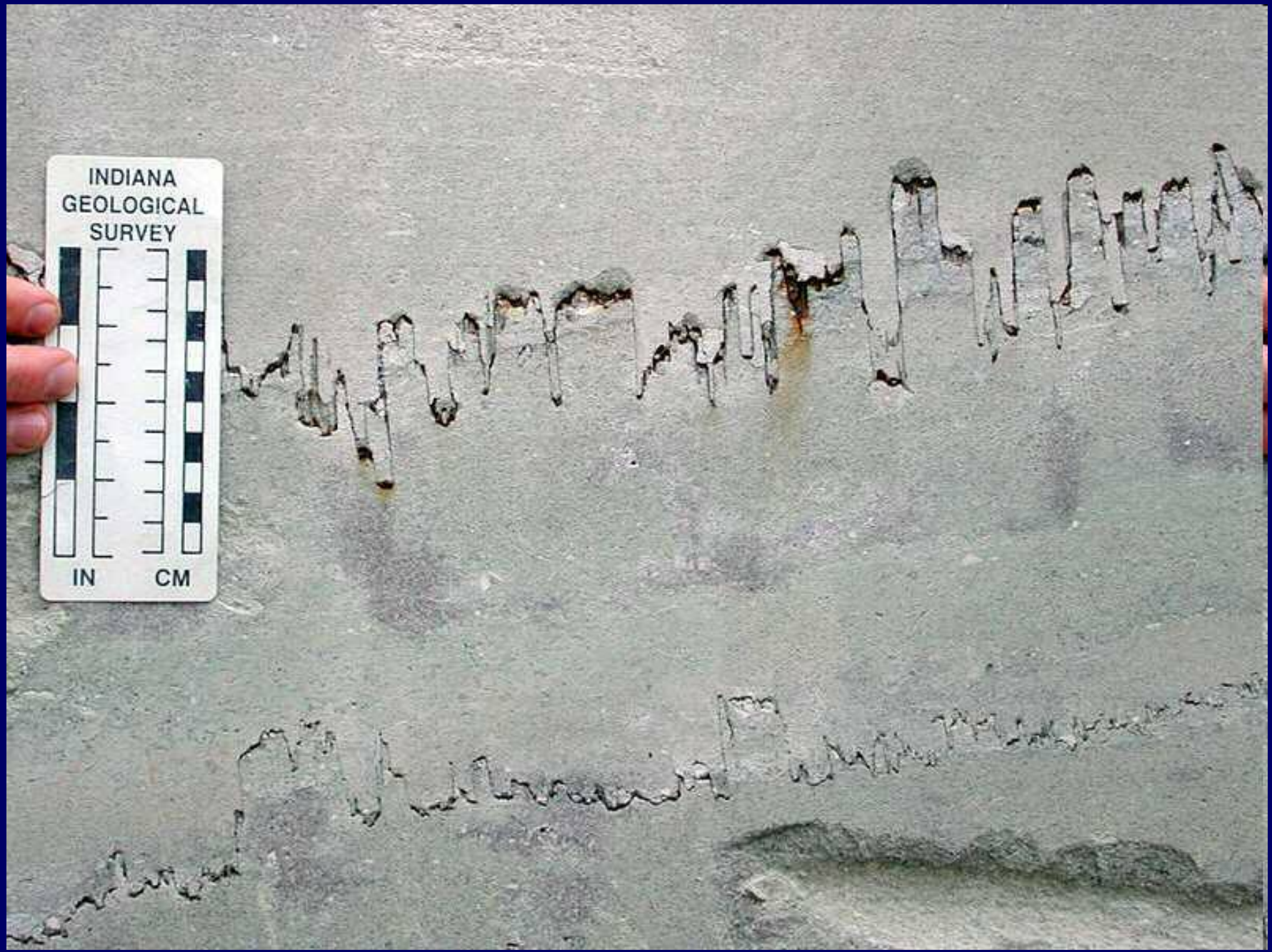
Compaction occurs as sediment is gradually buried

- re-packing of grains,
- breakage of weak grains
- water expelled

Results in porosity reduction and volume loss through grain repacking and pressure dissolution



Grain contact pressure dissolution




Pressure solution in limestone: formation of ‘stylolites’



Pressure solution in limestone evident from ammonites

# Dissolution and porosity increase

Calcite is soluble in surface waters, therefore calcite cemented sandstones often have their cement partially dissolved. Dissolution of the calcite cement results in secondary porosity



Mouldic porosity from solution of a gastropod shell

# Diagenetic processes: Formation of Dolomite

In carbonate rocks a diagenetic mineral (dolomite) is frequently encountered:

- double carbonate of calcium and magnesium
- $CaMg(CO_3)_2$
- forms by replacement of limestone:

$$2CaCO_3 + Mg^{2+} = CaMg(CO_3)_2 + Ca^{2+}$$

# Recognition of dolomite

Acid reaction: does not visibly react with cold dilute hydrochloric acid

Crystal shape: forms rhombohedral crystals

(Calcite in limestones is rarely rhombohedral)



# The puzzle of the origin of dolomite

Origin of dolomite is subject of much argument:

Common in rock record, rare in modern environment, difficult to create in laboratory using natural water under sedimentary-diagenetic conditions

Increase in abundance back in time- why?

- different composition sea water?
- dolomitization environments more prevalent?
- limestones have more time to get dolomitized?

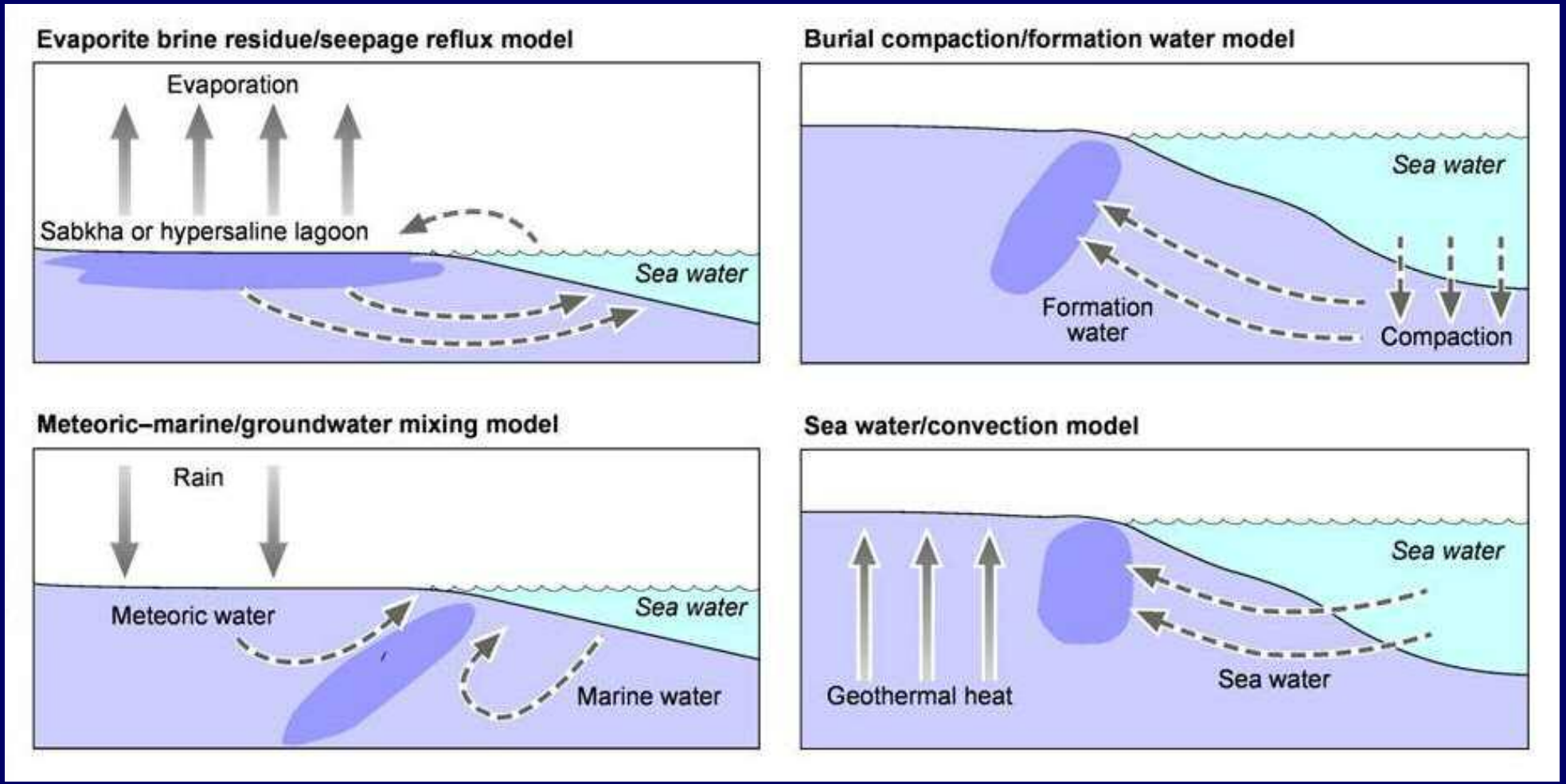
Required conditions:

Need Mg-rich fluid to flow through limestone

Seawater is only large reservoir of Mg ions (there is five times as much Mg as Ca in seawater)

Dolomitization more effective if water is warmed

# Four proposed models for dolomitization




(from Nicholls, 2009)

# Suggested Reading

Nicholls, 2009, 2nd edition
pp.225-246, 258-262, 279-291

If you would like a more thorough treatment:
Tucker, 2001, 3rd edition, chapter 3 and p. 110-130

# Revision from Year 1

# Other Cretaceous pelagic sedimentary rocks: Cherts





Well bedded primary chert



Secondary chert e.g., flint nodules in chalk

Key characteristics:

- very hard
- finely crystalline quartz
- formed by solution precipitation
- often slightly glassy in appearance
- conchoidal fracture

# Organisms whose hard parts are made of silica

- Organisms who secrete exoskeletons as amorphous silica = opaline silica $SiO_2.nH_2O$
- Two main groups – diatoms (Cretaceous onwards) and radiolaria (early Cambrian onwards)
- Diatoms – phytoplankton (floating photosynthetic algae) – mainly freshwater, some high latitude marine
- Radiolaria – zooplankton (microscopic animals with planktonic lifestyle), common in equatorial seas






Precambrian BIFs: alternating chert and iron oxide bands
This chert ppt directly from supersaturated sea water, not organisms



# Formation of siliceous oozes and primary cherts



Typical sequence of radiolarian cherts

Laminae in siliceous oozes reflect seasonal variation in plankton productivity and/or seasonal influxes of detrital clay

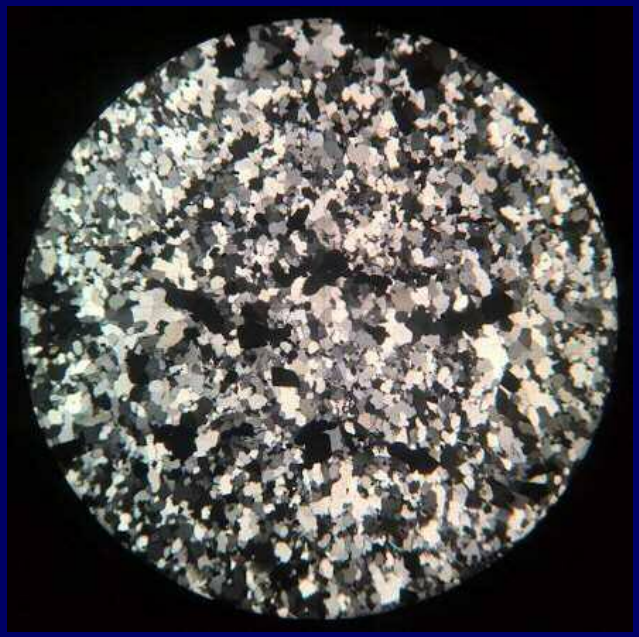
Why is it red?

Radiolaria and diatoms live in surface waters of ocean and sink to sea floor after death

Once incorporated in sediment and buried, the radiolaria/diatoms will slowly re-crystallize to quartz

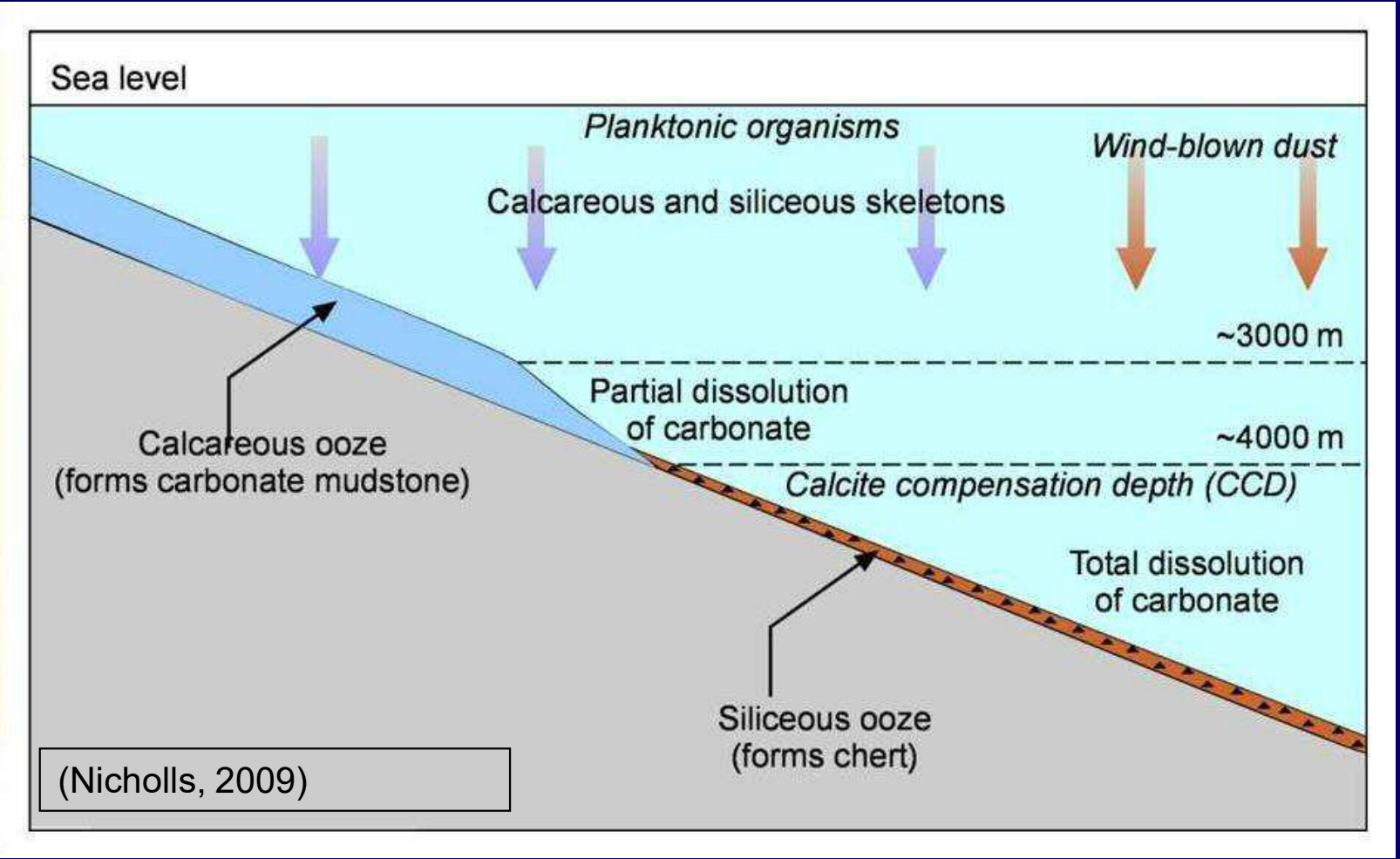
This takes a long time, typically tens of millions of years

Eventual product is a bedded chert



Thin section of cherts: note you can't see the individual radiolaria or diatoms as they have been recrystallised

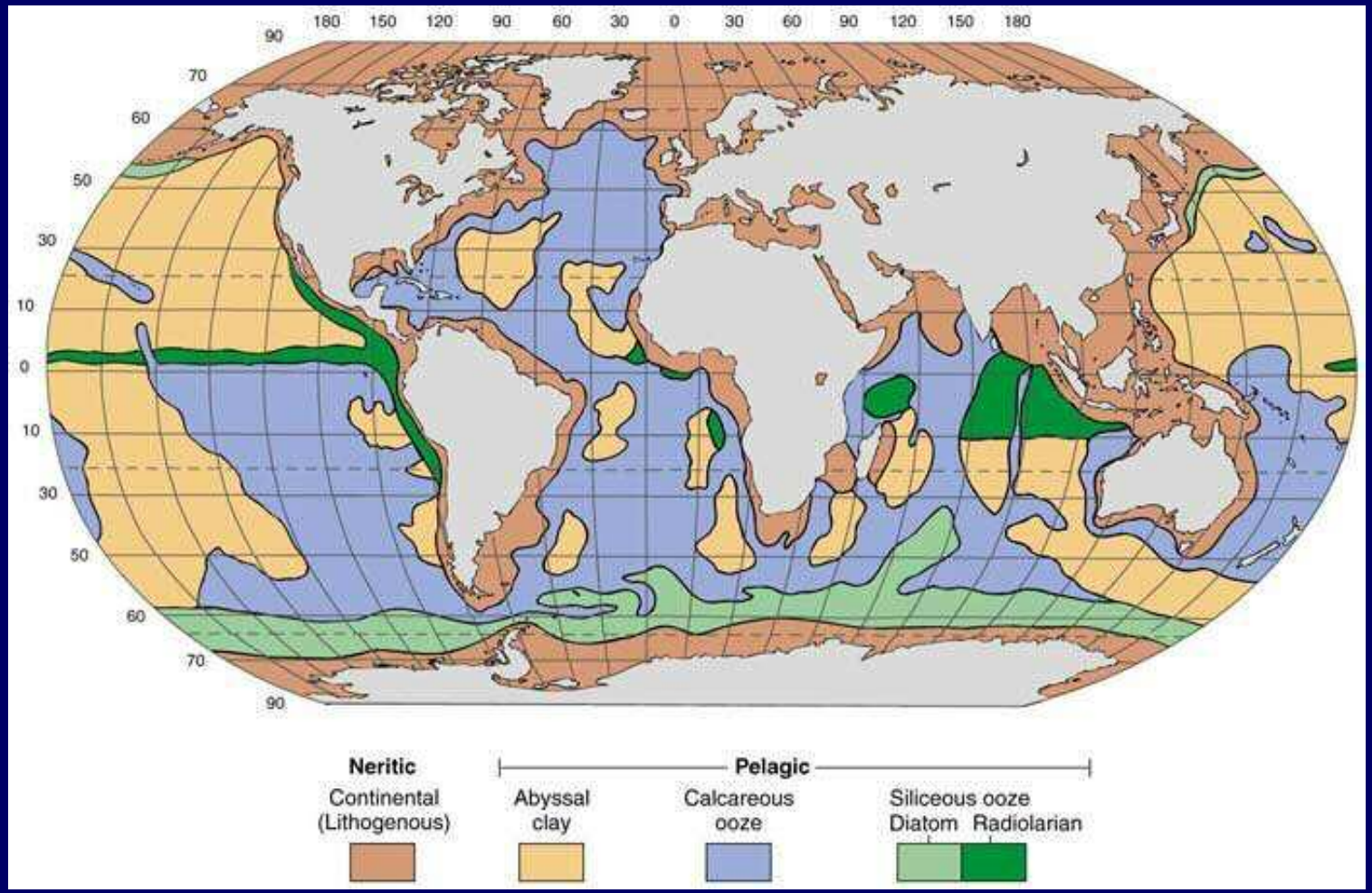
# Calcite compensation depth




## The CCD controls the distribution of siliceous and calcareous oozes on the ocean floor

- $CaCO_3$ solubility increases with depth (due to temperature decrease and pressure increase)
- At CCD rate of calcium carbonate material forming and sinking is equal to the rate the material is dissolving
- Most opaline silica dissolves in surface waters (warmer, Al/Si of shells, high surface area) only about 3% of global silica flux is buried in sediment

# The depth of the CCD has changed through time




Distribution of pelagic and neritic sediments today

- Cretaceous - early Eocene the CCD was much shallower due to intense volcanic activity = higher atmospheric $CO_2$ concentrations (higher $pCO_2$ in oceans) → increased dissolved $CO_2$ in surface waters
- Palaeozoic and early Mesozoic, prior to evolution of coccolithophores, siliceous oozes may have deposited at much shallower depths than when there was calcareous plankton available for sedimentation